Transfer task: defend and revise

antocciphysics | TEACHER KEY | CPA

Learning target: Defend the prediction with evidence and make a visible, justified revision.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 19; 3 course days later.*
The student multiplied by mass. Divide: $a=F_{net}/m=5$ m/s². Check units N/kg=m/s² or substitute ma back into F_{net}.

02. Explain a worked example (worked)
Prediction outside interval; possible constant velocity, start position, timing or measurement calibration needs testing.

03. Show the prediction (model)
Central (0,-120)→(16, 11.36). Lower (0,-120.05)→(16, 8.27); upper (0,-119.95)→(16, 14.45). Units million km, millions.

04. Write a public briefing (practice)
Use CPA end point (11.36, 35.20) million km; ranges x 8.27-14.45, y 34.4-36 million km. No real impact conclusion from this model.

05. Defend and revise (reason)
Evidence-specific feedback and revised calculation/label/reasoning. Accept defended unchanged step with physics justification; no invented peer feedback.

06. Choose a method (mixed) *Review: Unit 1, Day 10; 12 course days later.*
Earth pulls the sign down with 60 N; the cord pulls it up with 60 N.

07. Exit check (exit)
Require actual work references. Physically valid claim plus gravity/orbit question. Ratings are self-assessment only.

08. STRETCH: Apply it another way (stretch)
v=2.5 m/s; predict e.g. 15 m at 6 s and measure. Fitting one observation does not independently validate the model.

Teaching note: Independent transfer task, Day 2 of 2. Retain Day 21 work and submit it with this sheet. Use student reflection separately from the teacher's mastery rating.

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01. Bring back an earlier idea (retrieve) *Review: Unit 1, Day 19; 3 course days later.*
Central 38 m; half-width 4.5 m; interval [33.5, 42.5] m. Valid checks include units and upper/lower endpoint substitution.

02. Explain a worked example (worked)
Prediction outside interval; possible constant velocity, start position, timing or measurement calibration needs testing.

03. Graph central and bounded positions (model)
Central start -120 million km, end 11.365912 million km. Bounds initial±.05 million km; final half-width 3.140963 million km. Width grows with speed uncertainty.

04. Defend the public claim (practice)
Use Day 21 Honors answers. No impact probability inferred; gravity, Earth motion and 3 D state excluded. Explain bounds rather than present false certainty.

05. Peer defense and revision (reason)
Require specific feedback and justified revision or retained claim. Mathematical and physical reasoning must remain student’sown.

06. Choose a method (mixed) *Review: Unit 1, Day 10; 12 course days later.*
Weight 49 N down from Earth and tension 49 N up from string. Object-on-string acts on the string, not the object.

07. Exit check (exit)
Specific references in each dimension; revision with support; next question about gravity/orbits. Model does not establish real encounter.

08. STRETCH: Apply it another way (stretch)
Constant v predicts 15 m, inconsistent. One possible model: 2.5 m/s for first 4 s, then 1.5 m/s for next 2 s. More data needed to distinguish abrupt versus gradual changes.

Teaching note: Independent transfer task, Day 2 of 2. Retain Day 21 work and submit it with this sheet. Use student reflection separately from the teacher’s mastery rating.